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# Bidirectional purification of nonlinear guided waves using metamaterial filters for structural health monitoring

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ABSTRACT

Nonlinear guided waves (NGWs), owing to their exceptional sensitivity to microstructural variations in materials, have emerged as a powerful approach for detecting early-stage damage in structural health monitoring (SHM). However, such microstructural changes are typically very subtle, and the resulting nonlinear wave components are easily masked or distorted by other non-damage-related nonlinear sources within the system. Moreover, since the nonlinear energy is much weaker than that of the fundamental wave, existing approaches generally rely on frequency domain analysis to extract nonlinear components. This makes signal processing highly complex and imposes stringent requirements on the signal-to-noise ratio (SNR), thereby limiting the feasibility of real-time and reliable monitoring. To address these challenges, we propose a guided-wave purification strategy based on metamaterial filters (MFs). Bandgap-type MFs are attached to both the excitation and reception ends of the structure under inspection, enabling purification at both ends. These filters operate through local resonances induced by the coupling between patches and the host structure, thereby generating bandgaps within the target frequency range and effectively blocking the propagation of specific frequency components. The excitation-side filter suppresses parasitic second harmonic components generated by the excitation system, thereby eliminating unwanted nonlinear interference. Meanwhile, the reception-side filter enables selective extraction of damage-induced second harmonic components, allowing real-time identification of the structural health condition through variations in the second harmonic time domain signal. Finite element simulations and laboratory experiments conducted on aluminum plates demonstrate that the proposed bidirectional purification strategy can effectively suppress nonlinear interference and extract damage-related nonlinear responses, thereby simplifying signal processing procedures and exhibiting promising potential for practical engineering applications.

## 1. Introduction

In critical infrastructures such as aerospace, wind energy, and rail transportation, structural health monitoring (SHM) has become an indispensable approach to ensuring operational safety and extending structural service life. Ultrasonic guided-waves based nondestructive testing has attracted significant attention in recent years, owing to its long propagation distance and high sensitivity to interfacial debonding and early-stage damage [1,2]. Compared with conventional point-to-point inspection methods, guided waves can interrogate large areas of a structure, making them particularly suitable for in-service monitoring of large-scale and complex structures. Linear guided waves have been proven effective for detecting macroscopic defects based on scattering characteristics and are widely used in SHM [3,4]. However, their ability to identify early-stage micro-damage is limited, particularly when the defect is too small to produce significant scattering or reflection, resulting in reduced sensitivity [5,6]. In contrast, nonlinear guided waves (NGWs) exhibit a much stronger response to local material and geometric inhomogeneities, such as microcracks, localized contacts, and stress concentrations during propagation. Energy can be transferred from the fundamental frequency to higher harmonics, generating nonlinear components such as second harmonics [7–9], thereby offering richer and more damage-sensitive information [10–13].

Although nonlinear ultrasonics are theoretically endowed with high

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sensitivity and selectivity, their practical application in engineering remains fraught with challenges. First, the nonlinear effects induced by micro-damage are usually extremely weak, and the resulting nonlinear energy is easily obscured by system nonlinearities such as those arising from excitation devices or boundary contacts [14,15]. Second, because these nonlinear components are two orders of magnitude weaker than the fundamental wave, current extraction methods predominantly rely on frequency or time–frequency domain analyses such as Fourier transform, frequency–phase modulation, or wavelet packet decomposition [16–18]. However, these techniques are highly sensitive to SNR and parameter settings, involve complex processing workflows, and are largely confined to post-processing, without actively suppressing nonlinear interference along the propagation path or enabling real-time health assessment directly from time domain signals. Therefore, achieving effective purification of guided-wave propagation from a physical perspective—so that damage-induced nonlinear responses are not masked by nonlinear interference originating from the excitation system, and nonlinear features can be directly extracted at the monitoring point with reduced reliance on complex post-processing—has become a critical challenge for improving both the usability and real-time applicability of NGWs. This need motivates the present work, which aims to actively regulate nonlinear components during wave propagation through structural tuning mechanisms. Specifically, under the premise of preserving the structural integrity of the specimen, nonlinear interference introduced by system sources is suppressed at the excitation end to purify the input wavefield, while the dominant fundamental component is filtered at the reception end to enable the extraction of nonlinear features. This strategy is expected to enable more intuitive and robust evaluation of structural conditions, thereby significantly improving the practicality and engineering adaptability of NGWs for SHM. In recent years, the advent and rapid development of metamaterials have opened new avenues for this approach, making the purification and active control of guided-wave signals feasible from a physical perspective.

Metamaterials are artificially engineered structures composed of periodic unit cells (UCs), whose macroscopic responses are governed by the geometry and arrangement of the units rather than by the intrinsic physical properties of the base material. Through careful geometric design of the UCs, elastic wave propagation can be manipulated at subwavelength scales [19–27], enabling a wide range of unconventional phenomena such as bandgaps, negative effective mass density and anisotropic transmission. These unique capabilities have spurred extensive research into elastic-wave control, encompassing applications in waveguiding design [19], vibration isolation [20–22], energy focusing [23,24], and topological transport [25–27]. Despite the rapid development of metamaterials for linear guided-wave manipulation, explorations of their application in NGWs–SHM remain relatively scarce. Recent studies aimed at enhancing the sensitivity and robustness of nonlinear guided-wave detection can be broadly classified into two categories: enhancement of nonlinear effects and physical purification of harmonic components. Miniaci et al. introduced bandgap phononic crystals in aluminum plates to selectively enhance higher harmonics, thereby improving second harmonic sensitivity to microcracks and other nonlinear sources [28]. From a different perspective, Tian et al., Liu at al. and Yi et al. exploited metamaterial-enabled mode conversion to preferentially excite propagation modes that support stronger nonlinear accumulation [29–32]. More recently, Tian et al. further demonstrated defect-mode elastic metamaterials for localized amplification of nonlinear ultrasonic signatures in fatigue-damage detection [33]. In addition, the physical purification of nonlinear guided-wave signals through metamaterial filtering has attracted increasing attention. Early studies by Shan et al. demonstrated that bandgap-based MFs can suppress spurious harmonics in NGWs [34]. Subsequently, topology-optimized metamaterial structures [35] and 3D-printed phononic-crystal ultrasonic filters [36–38] were introduced to physically separate fundamental and harmonic components during wave propagation, thereby improving the reliability of nonlinear ultrasonic measurements. On this basis, metamaterial filtering has also been applied to nonlinear guided-wave damage localization. Representative works demonstrated that embedded or additively manufactured phononic-crystal filters can suppress spurious harmonics and thereby improve the spatial accuracy and robustness of nonlinear source localization in SHM systems [39–41]. Furthermore, Gao *et al.* generalized the harmonic purification concept to characteristic guided waves, indicating the broader applicability of metamaterial-based filtering strategies for suppressing nonlinear interference in complex guided-wave SHM systems [42]. Nevertheless, these applications still face notable limitations. Their structural designs are generally complex and impose stringent installation requirements, and they often rely on sophisticated data-processing procedures to infer structural health from frequency domain features, which makes it difficult to meet the real-time and engineering adaptability demands of SHM.

To address the challenges of weak second harmonic responses, susceptibility to interference, and the complexity of signal post-processing that hinder real-time monitoring in NGWs, We propose a nonlinear guided-wave purification strategy based on MFs, specifically designed for SHM. MFs is composed of periodically arranged UCs, defined as the combination of triangular patches and the host structure. In each UC, the patches are symmetrically bonded to the upper and lower surfaces of the specimen, and together with the substrate they form an integrated system that exhibits bandgap characteristics. We systematically investigated the effects of patch geometry and material properties on bandgap characteristics, thereby elucidating the customizable frequency-filtering capability of the MFs and its underlying physical mechanisms. Building on this concept, a pair of MFs tuned to the designed bandgaps are integrated at the excitation and reception ends to realize bidirectional purification of NGWs. The excitation-side filter suppresses spurious harmonics generated in the excitation system, while the reception-side filter selectively extracts damage-related second harmonic components by filtering out the dominant fundamental wave. Numerical simulations and laboratory experiments consistently verify that the proposed bidirectional purification strategy effectively mitigates system-induced nonlinear interference, simplifies signal-processing procedures, and enhances the stability and detectability of micro-damage. These results establish a feasible and robust pathway for the practical deployment of NGWs in complex SHM applications and provide a solid physical foundation for metamaterial-assisted SHM technologies.

The rest of this paper is organized as follows. Section 2 introduces the design of the MFs, including dispersion curve analysis and frequency domain analysis. Section 3 presents the time domain finite element analysis to validate the proposed approach. Section 4 provides the experimental validation results. Finally, conclusions are presented in Section 5.

## 2. Metamaterial filters design

### 2.1. Unit cells design and desired band feature

In ultrasonic SHM, NGWs provides high sensitivity for assessing structural health, enabling the effective detection of subtle defects such as microcracks, interfacial contacts, and localized plastic strains. When waves propagate through a structure and interact with these micro-damages, part of the energy is transferred from the fundamental excitation frequency $f$ to its integer multiples, thereby generating harmonic components. Among these, the second harmonic $2f$ is the most common and readily detectable nonlinear signature. Owing to the shorter wavelength of higher order harmonics, smaller defects can be detected, making them a crucial physical indicator for identifying early-stage micro-damage.

Fig. 1 illustrates the system framework of the proposed nonlinear guided-wave bidirectional purification strategy based on MFs. As shown in Fig. 1(a), without the MFs the system configuration is identical to that

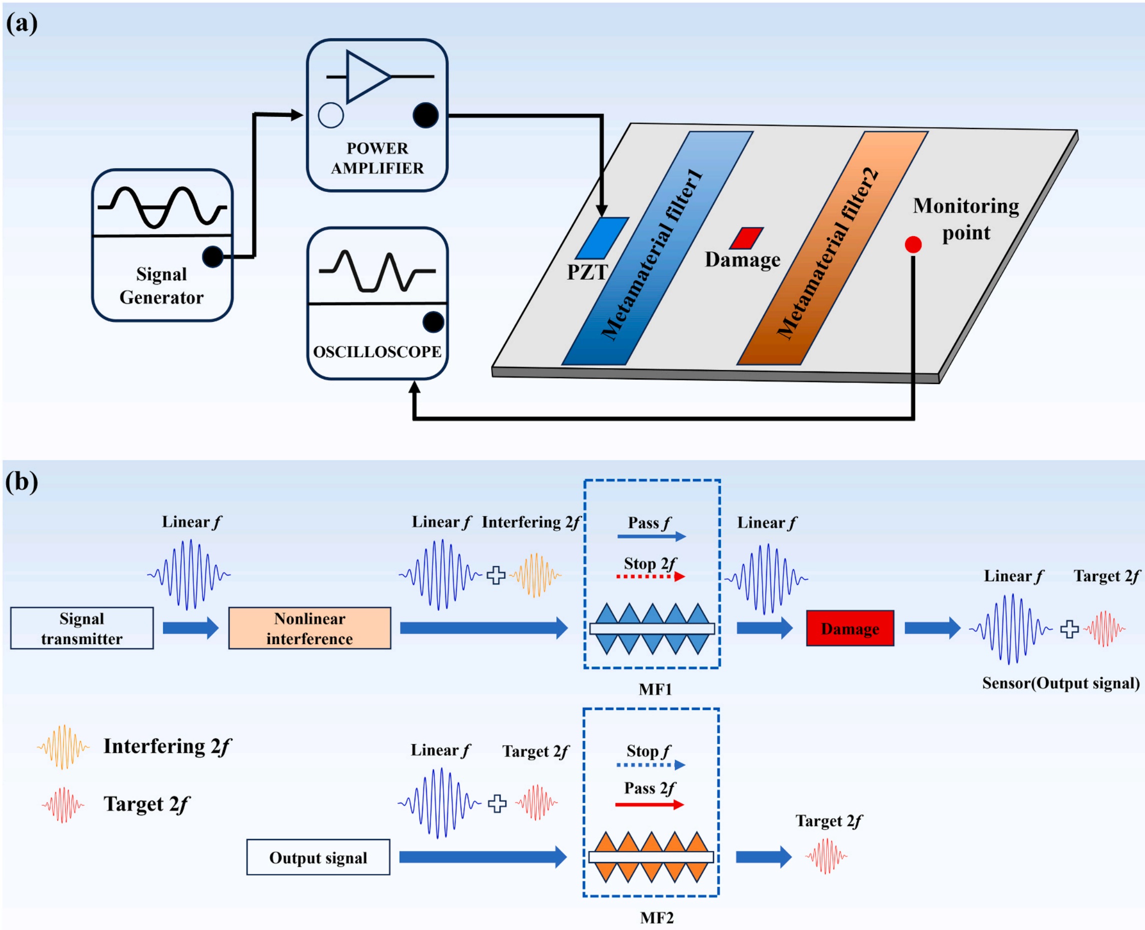


**Fig. 1.** (a) Schematic of nonlinear guided-wave bidirectional purification using MFs; (b) Signal flow diagram of NGWs under the action of the MFs.

of a typical SHM scheme; with the MFs incorporated, bidirectional purification of the guided-wave signals can be achieved. The signal-purification process is depicted in Fig. 1(b). A 180 kHz fundamental mode signal (blue) is generated by the excitation chain comprising a signal generator, power amplifier, and PZT actuator. However, nonlinear interference may arise from device aging or imperfect PZT bonding, producing spurious second harmonic signals (orange). After passing through MF1, these spurious harmonics are effectively suppressed, leaving only the fundamental signal (blue). When the guided wave propagates through a damaged region, the desired nonlinear response appears in the form of a second harmonic (red). The fundamental wave carrying the damage-induced second harmonic then passes through MF2, which blocks the fundamental component while transmitting only the second harmonic. This enables direct assessment of structural health from time domain signals, without the need for complex frequency domain processing.

Considering an aluminum plate with a thickness of $t = 2$ mm as the specimen for detection, the dispersion curves of both phase velocity and group velocity are calculated, as shown in Fig. 2. Previous studies have demonstrated that only under conditions of nonzero power flux and phase velocity matching can the fundamental wave and its harmonics remain phase-synchronized during propagation, thereby enabling

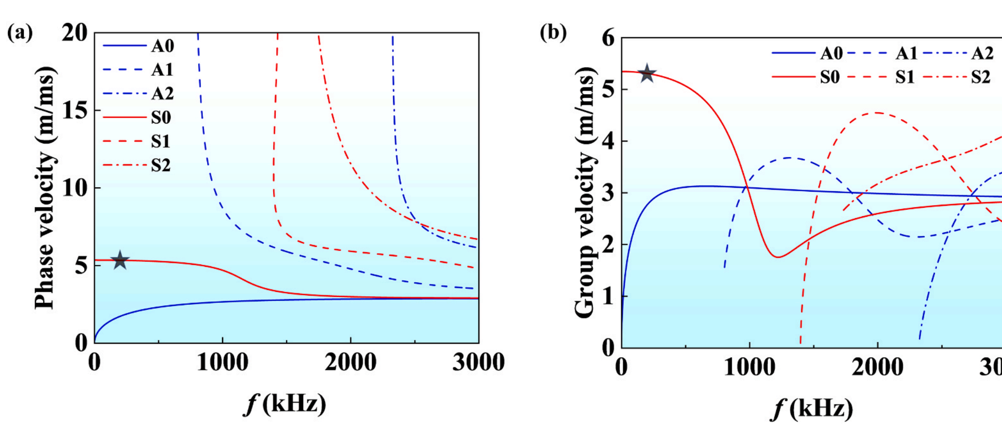


**Fig. 2.** Dispersion curves of the 2 mm aluminum plate, (a) phase velocity; (b) group velocity.

continuous energy transfer into nonlinear components. Therefore, an excitation frequency of 180 kHz (marked by the black star) was selected to monitor the aluminum plate and to validate the proposed bidirectional purification strategy. At this frequency, the S0 mode exhibits favorable phase-velocity matching with its second harmonic, allowing nonlinear energy to accumulate spatially along the propagation path, which enhances the detectability of early-stage micro-damage. In addition, only the fundamental S0 and A0 modes exist at 180 kHz, thereby minimizing modal complexity during guided-wave propagation.

In the context of SHM, the designed metamaterials are required to achieve bidirectional purification of guided waves without compromising the integrity of the specimen. To enhance structural robustness, the configuration should also be as simple as possible. Based on this principle, this study adopts the design concept of generating bandgaps through periodic UCs. Triangular patches are periodically bonded to the upper and lower surfaces of the aluminum plate to construct a MF with a specific frequency bandgap, as shown in Fig. 3(a). To implement the proposed bidirectional guided-wave purification strategy, two types of UCs are further designed and assembled to form two MFs. Each MF selectively filters different frequency components, thereby enabling effective control of nonlinear interference at the excitation end and targeted nonlinear features at the reception end. Given the excitation frequency, UC1 is designed to create a bandgap that covers the second harmonic frequency at 360 kHz while allowing the fundamental frequency at 180 kHz to pass through. In contrast, UC2 is designed to create a bandgap at 180 kHz while ensuring the passage of the second harmonic at 360 kHz. At the same time, the bandgaps of both UCs should be as wide as possible to maximize the purification capability of the MFs, thereby enabling more accurate and reliable damage detection in SHM applications.

### 2.2. Numerical simulation methods

Based on the design requirements for the UCs, two UCs are designed to form the MFs, as shown in Fig. 3(b). The geometrical parameters of the two UCs, including the lattice constants $a_1 = 4$ mm and $a_2 = 3.2$ mm, the triangular patch heights $h_1 = 1.5$ mm and $h_2 = 1$ mm, and the adhesive layer height $h_3 = 0.05$ mm, are summarized in Table 1. The patch colors correspond to different materials: blue triangles represent aluminum and orange triangles represent lead, which are used to construct the UCs of MF1 and MF2. The material properties for aluminum, lead, and the adhesive layer, including mass density $\rho$, elastic modulus $E$, and Poisson's ratio $\upsilon$, are provided in Table 2.

The dispersion curves of the designed UCs were calculated using the Solid Mechanics module in COMSOL Multiphysics. The eigenfrequency solver of the module was employed to study the bandgap characteristics of the UCs. In this simulation, the boundary conditions (BCs) and mesh settings are shown in Fig. 4(a) and (b). The left and right boundaries of the aluminum plate and adhesive layer were set with Bloch-Floquet BCs, while the adhesive layer was meshed using quadrilateral free meshing, and the aluminum plate along with the triangular patches were meshed with triangular free meshing. To calculate the dispersion relation, we scanned the first irreducible Brillouin zone of the UCs. Fig. 4(c) illustrates the process of scanning the Bloch wavevector $k$, from 0 to $\pi/a$ with a step size of $0.02\pi/a$. A total of 20 eigenfrequencies were computed. Additionally, to ensure the accuracy of the results, mesh convergence was carefully verified. The maximum element size of the mesh (*MEM*) was varied from $a/5$ to $a/100$, with a step size of $a/2$, where $a$ refers to the lattice constant. The upper and lower bandgap frequencies, as well as the degrees of freedom (DOFs) for each mesh size, are presented in Fig. 4(d) and (e). As the MEM decreases, the number of DOFs increases significantly, leading to longer computation times. Meanwhile, the bandgap frequencies gradually stabilize, indicating that the mesh has sufficiently converged. Based on a balance between computation time and accuracy, a mesh size of a/90 was chosen for calculating the bandgap characteristics of the UCs.

**Table 1**
The geometric parameters of the MFs.

| | $t$ | $a_1$ | $a_2$ | $h_1$ | $h_2$ | $h_3$ |
|---|---|---|---|---|---|---|
| Unit: mm | 2 | 4 | 3.2 | 1.5 | 1 | 0.05 |

**Table 2**
Material parameters.

| | $\rho$ (kgm$^{-3}$) | $E$ (GPa) | $\upsilon$ | $\bar{A}$ (GPa) | $\bar{B}$ (GPa) | $\bar{C}$ (GPa) |
|---|---|---|---|---|---|---|
| Aluminum | 2700 | 70 | 0.33 | -351.2 | -140.4 | -102.8 |
| Adhesive | 1080 | 1.31 | 0.4 | -20.9 | -8.3 | -6.1 |
| Pb | 11370 | 16 | 0.42 | - | - | - |

To validate the bandgap characteristics of the UCs, a frequency domain finite element model was established, as shown in Fig. 4(f). The model was used to evaluate the linear filtering effect of the MFs and calculate their transmission coefficient ($T_c$). Ten UCs were used to form the MFs. A boundary excitation was applied at the leftmost edge of the aluminum plate, with a displacement of 1 mm in the $x$-direction. The lengths of the aluminum plates before and after the MFs were defined as $l_1$ and $l_2$, both 200 mm. To reduce boundary reflection effects, a perfectly matched layer (PML) of length $l_3 = 50$ mm was placed at the far end. Then, the frequency domain analysis was conducted in COMSOL Multiphysics with a frequency scan from 5 kHz to 1000 kHz at a step size of 1 kHz. This frequency scan allows for a comprehensive evaluation of the filtering capabilities of the MFs. To quantify the filtering effect of the designed MFs, The $T_c$ is defined as

$$T_c = 20\log\frac{u_2}{u_1} \tag{1}$$

Where $u_1$ represents the integral of the square of the in-plane displacement over the region $S_1$ before the MFs in Fig. 4(f), and $u_2$ represents the integral of the square of the in-plane displacement over the region $S_2$ after passing through the MFs. These are expressed as:

$$u_1 = \iint_{S_1} |u_x|\, dS_1 \tag{2}$$

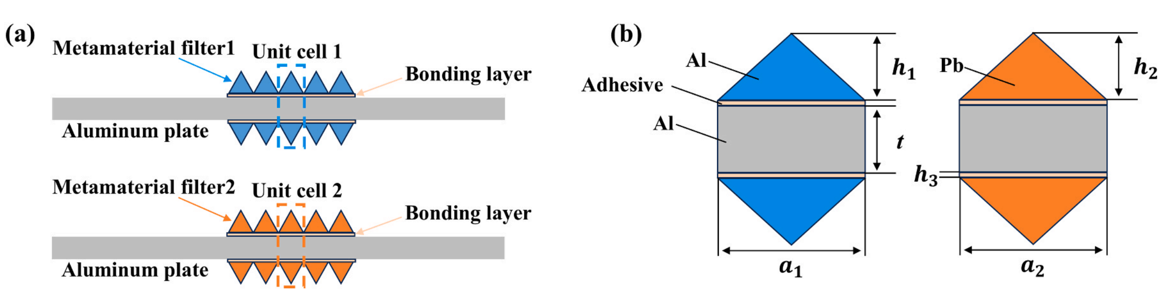


**Fig. 3.** (a) Schematic layout of the bidirectional MFs; (b) Configuration of the UCs of the MFs.

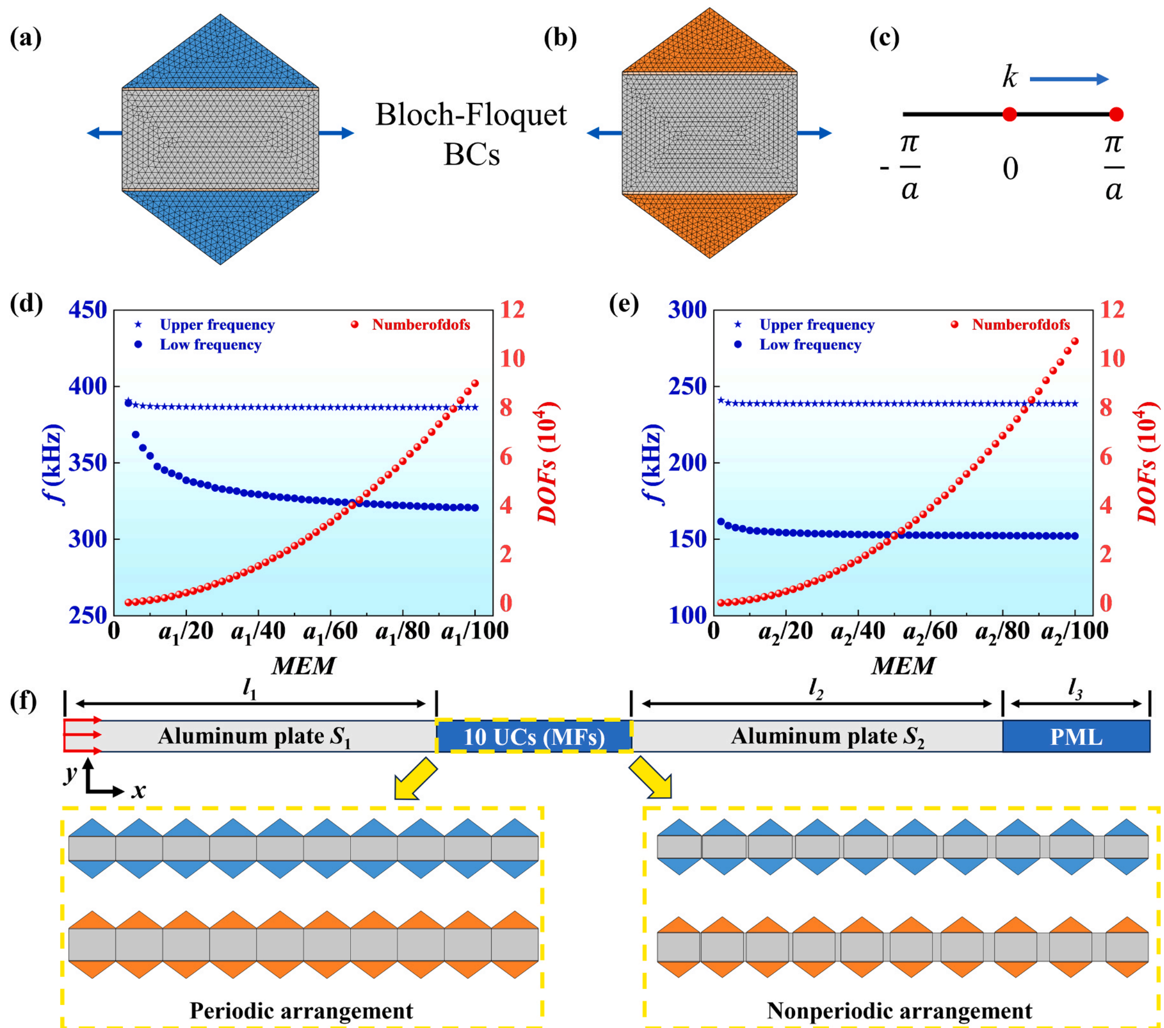


**Fig. 4.** Dispersion curves calculation and mesh convergence analysis of the UCs, along with the BCs setup for the frequency domain analysis. (a, b) BCs for UC1 and UC2, respectively; (c) Bloch wavevector scanning path within the first Brillouin zone; (d, e) Convergence of the upper and lower bandgap frequencies with respect to mesh size for UC1 and UC2, respectively; (f) BCs setup for the finite element models used in the frequency domain analysis.

$$u_2 = \iint_{S_2} |u_x| \, dS_1 \tag{3}$$

Where $u_x$ is the in-plane displacement.

### 2.3. Bandgap characteristics and frequency domain analysis

Based on the validated simulation setup and mesh convergence analysis, the dispersion relations of the designed UCs were obtained, and the corresponding bandgap characteristics are shown in Fig. 5. Since the selected propagation mode in this work is the S0 mode, whose displacement primarily manifests as in-plane components, only in-plane displacement is considered during the analysis. The mode type can be determined by introducing the polarization factor $\alpha$, where $\alpha = 0$ corresponds to out-of-plane modes and $\alpha = 1$ corresponds to in-plane modes. This can be mathematically expressed as:

$$\alpha = \iint_{UC} \frac{|u_x|^2}{(|u_x|^2 + |u_y|^2)} dS (0 \le \alpha \le 1) \tag{4}$$

For visualization, the two types of modes are distinguished in the band diagrams, with out-of-plane modes shown in blue and in-plane modes shown in yellow. Fig. 5(a) shows the dispersion curves of UC1, where a complete polarization bandgap appears in the range of 320–390 kHz (blue shaded region), preventing in-plane displacements from propagating through MF1. In contrast, a pass band is observed near 180 kHz, allowing in-plane displacements to pass through. To further confirm the bandgap effect, the $T_c$ of MF1 composed of 10 UC1s was calculated. As shown in Fig. 5(b), the yellow curve exhibits a sharp decline near 360 kHz, consistent with the bandgap in the dispersion curves while the $T_c$ remains close to 1 at 180 kHz, corresponding to a pass band. This demonstrates that MF1 can suppress the second harmonic interference while maintaining high transmission of the fundamental frequency.

To investigate the mechanism of bandgap formation, modal and structural analyses were performed. Previous studies indicate that bandgaps mainly arise from two mechanisms: local resonance or Bragg scattering [43–47]. Local resonance isolates energy through UC resonances and blocks propagation, being insensitive to spatial periodicity, whereas Bragg scattering results from multiple scattering and interference caused by periodic structures, requiring spatial periodicity comparable to the wavelength to achieve bandgap characteristics. In our case, based on the dispersion relation of the 2 mm aluminum plate, the S0 mode wavelength at the bandgap frequency is $\lambda = 29.6$ mm and the lattice constant of UC1 is $a_1 = 4$mm, yielding $\lambda/a_1 = 7.4$, i.e., a deep-subwavelength regime, which is more consistent with a locally resonant mechanism. Fig. 5(c) and 5(d) show the structural deformations at frequency points A and B in the dispersion curve. Near the bandgap frequency, energy is concentrated in the triangular patches on

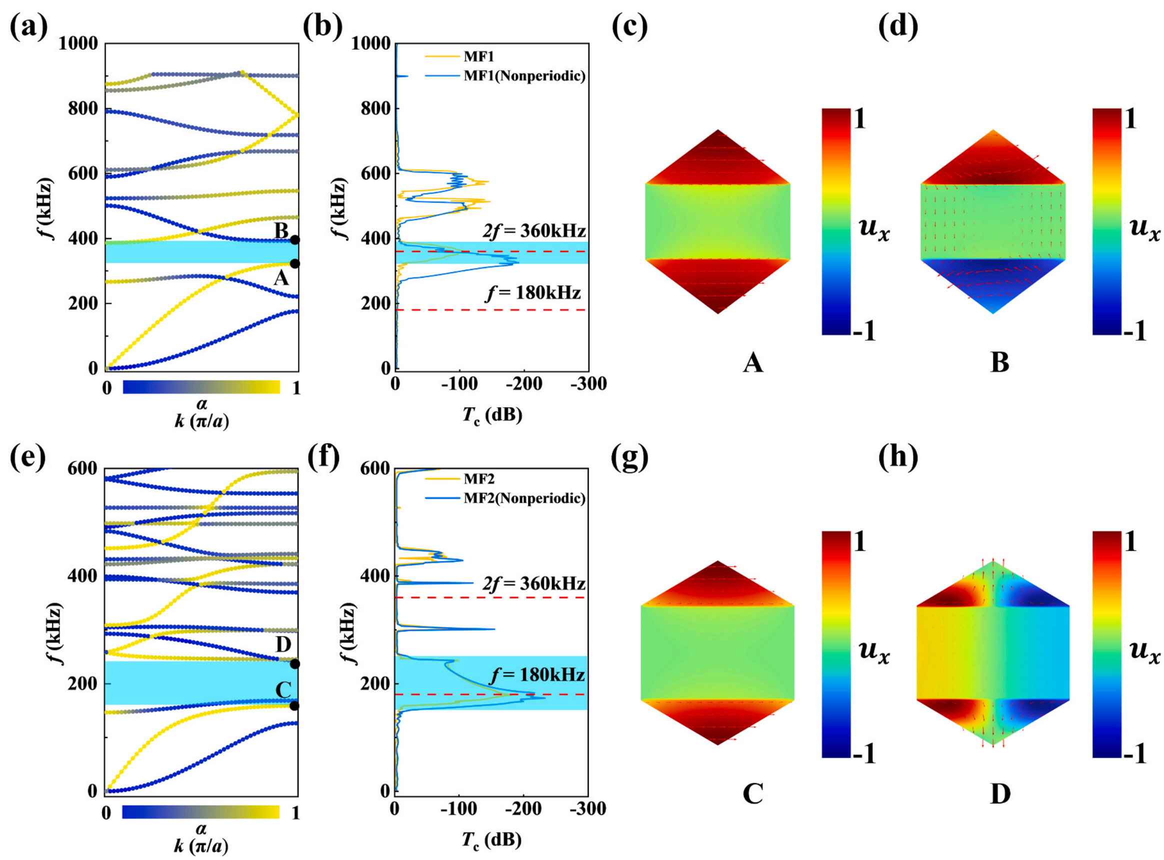


**Fig. 5.** Dispersion curves and mode analysis of the UCs and the $T_c$ of MFs. (a) Dispersion relation of UC1; (b) Comparison of $T_c$ between MF1 composed of 10 periodically arranged UC1s and a nonperiodic arrangement of 10 UC1s; (c, d) Structural deformation corresponding to frequency points A and B in panel (a); (e) Dispersion relation of UC2; (f) Comparison of $T_c$ between MF2 composed of 10 periodically arranged UC2s and a nonperiodic arrangement of 10 UC2s; (g, h) Structural deformation at frequency points C and D in (e).

the top and bottom surfaces, exhibiting typical local resonance features. Furthermore, as illustrated in Fig. 4(f), the periodicity of the 10 UC1s array was disrupted by introducing gradient spacing, and the corresponding $T_c$ was recalculated, represented by the blue curve in Fig. 5(b). The bandgap persisted and filtering performance was even enhanced, further confirming that the bandgap of MF1 originates from local resonance rather than Bragg scattering.

Fig. 5(e) – (h) show the analysis of the reception side filter MF2 ($\lambda/a_2$ = 9.25). In contrast to MF1, MF2 exhibits a complete polarization bandgap within 167–245 kHz and an in-plane polarization bandgap within 155 – 167 kHz, where in-plane displacements cannot pass through. Meanwhile, a stable pass band appears near 360 kHz. The $T_c$ of MF2 with 10 periodically arranged UCs (yellow curve in Fig. 5(f)) shows a sharp reduction near 180 kHz and nearly full transmission near the second harmonic. When periodicity was broken, the $T_c$ remained almost unchanged in the target band, confirming that the bandgap originates from local resonance rather than Bragg scattering. Fig. 5(g) and 5(h) correspond to the eigenmodes at points C and D in the dispersion curves, similar to the bandgap characteristics of UC1, showing energy concentrated in the triangular patches. The difference is that, near the upper bandgap frequencies of UC2, the energy is primarily concentrated at the two corners of the triangular patches. This bandgap mechanism, induced by local resonance, endows the filter with enhanced robustness: even if the patch positions inevitably deviate in practical applications, the local resonance bandgap remains stable, ensuring stable filtering performance.

To further validate the frequency selective capability of the designed MFs, frequency domain finite element simulations were performed using COMSOL Multiphysics, with the detailed settings shown in Fig. 4(f). The propagation of MF1 and MF2 was examined at the excitation frequency ($f$ = 180 kHz) and its second harmonic ($2f$ = 360 kHz). The results are shown in Fig. 6(a)-(h). Fig. 6(a) and 6(b) show the response of MF1. At 180 kHz, the incident guided wave propagates through the filter region, indicating that this frequency lies within the pass band of MF1. At 360 kHz, as the wave propagates through MF1, resonance between the aluminum plate and the triangular patches traps energy within the patches, thereby forming a bandgap that blocks wave transmission. This demonstrates that MF1 allows efficient transmission of the fundamental frequency while effectively suppressing the second harmonic. In addition, Fig. 6(c) and 6(d) show the results of MF1 in a nonperiodic arrangement. The energy trapping and bandgap formation still occur even without the periodic arrangement of the UC1s, which corresponds to the $T_c$ calculated in Fig. 5(b). This highlights the advantage of local resonance in MF1, where the filtering capability does not rely on the periodicity of the UCs, allowing for a robust filtering effect even in nonperiodic configurations. Similarly, Fig. 6(e) and 6(f) depict the selective transmission behavior of MF2. At 180 kHz, the incident wave is significantly attenuated within the filter region, confirming that the fundamental frequency falls within MF2's bandgap. In contrast, at 360 kHz, the guided wave propagates stably through MF2, indicating that the second harmonic lies within its pass band. Thus, MF2 effectively suppresses the fundamental frequency while allowing the second harmonics to pass through. Fig. 6(g) and 6(h) further illustrate the behavior of MF2 under nonperiodic conditions. Similar to MF1, even when the periodicity is disrupted, the selective filtering effect remains intact. The selective filtering results of MF1 and MF2 provide the foundation for our proposed bidirectional purification strategy.

To enable flexible tuning of the bandgap range for various

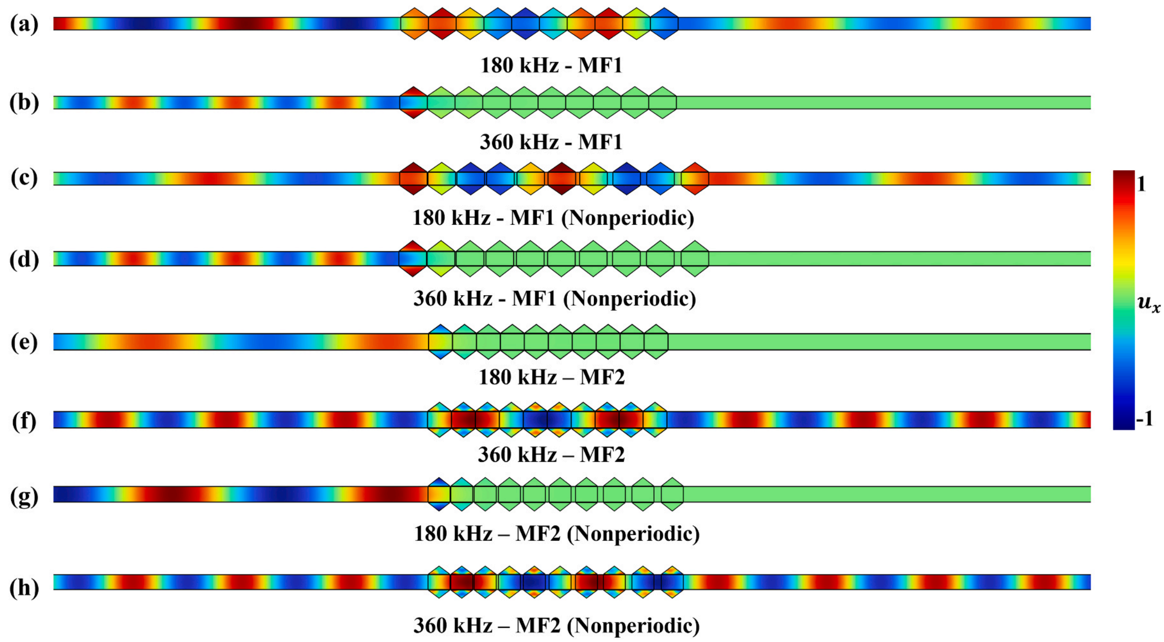


**Fig. 6.** Displacement fields at 180 kHz and 360 kHz with MF1 and MF2, for both with periodic and nonperiodic configurations. (a, b) Displacement fields at 180 kHz and 360 kHz with MF1; (c, d) MF1 in a nonperiodic arrangement; (e, f) Displacement fields at 180 kHz and 360 kHz with MF2; (g, h) MF2 in a nonperiodic arrangement.

monitoring scenarios, we systematically investigated the correlation between geometric structural parameters and bandgap characteristics. We focused on two key bandgap parameters: the bandgap size $B = f_u - f_l$ and its center frequency $\omega_c = (f_u + f_l)/2$, where $f_u$ and $f_l$ are the upper and lower frequency bound of bandgap. We performed parameter sweeps for the lattice constants ($a_1$ and $a_2$, ranging from 2.5 mm to

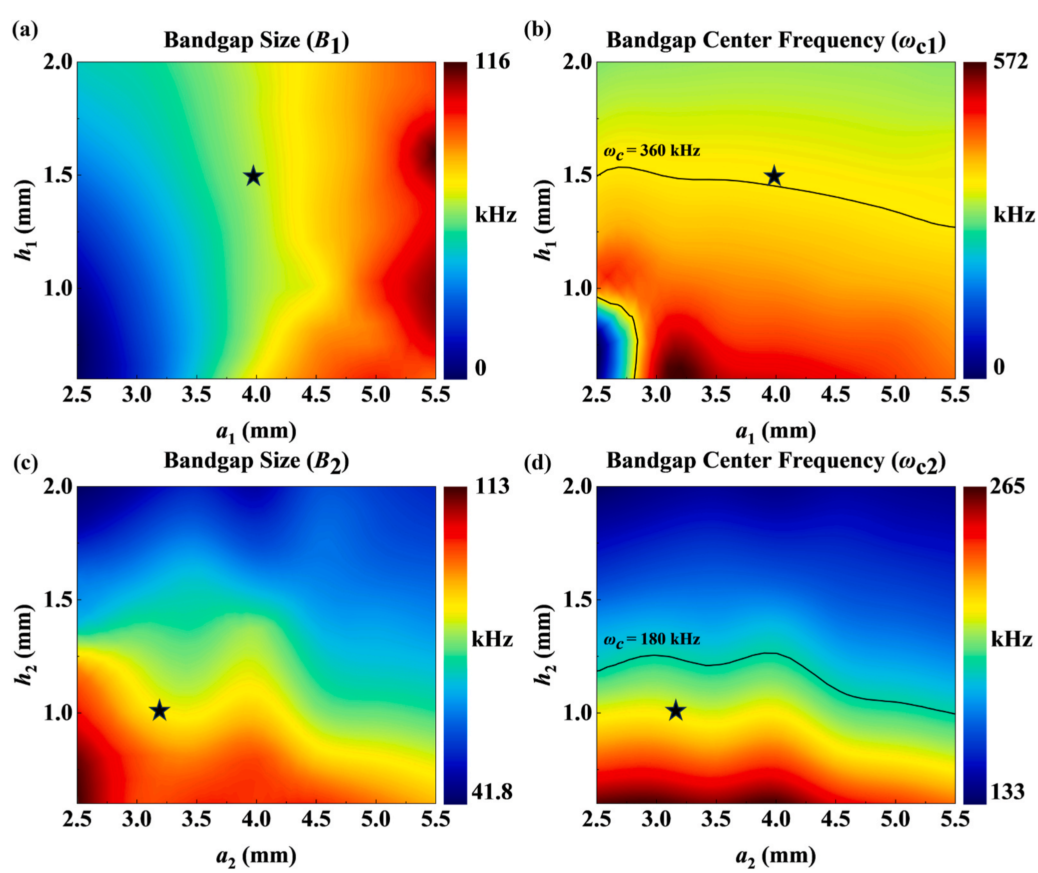


**Fig. 7.** Effects of design parameters on the bandgap characteristics of UC1 and UC2. (a-b) illustrate the influence of the lattice constant $a_1$ and the height $h_1$ of the triangular patches on the $B_1$ and $\omega_{c1}$ for UC1. (d-f) show how the lattice constant $a_2$ and the height $h_2$ of the patches affect $B_2$ and $\omega_{c2}$ for UC2.

5.5 mm with a step size of 0.1 mm) and the heights ($h_1$ and $h_2$, ranging from 0.5 mm to 2 mm with a step size of 0.1 mm), as shown in Fig. 7. Fig. 7(a) illustrates that $B_1$ expands with increasing $a_1$ and $h_1$. In contrast, Fig. 7(c) demonstrates that for UC2, the $B_2$ becomes broader as both $a_2$ and $h_2$ decrease. Regarding the center frequency, Fig. 7(b) and (d) demonstrate a consistent trend for UC1 and UC2: the $\omega_{c1}$ and $\omega_{c2}$ increases as $a_1$, $h_1$, $a_2$ and $h_2$ decrease. This is because smaller geometric dimensions reduce the UC's equivalent mass, raising the local resonant frequency of the patch-adhesive system. The contour lines in Fig. 7(b) and (d) mark the center frequencies of 360 kHz for UC1 and 180 kHz for UC2, with the design parameters selected in this study indicated by black pentagrams. It is worth noting that in selecting the parameters for UC2, we deviated from the center frequency of 180 kHz for two reasons: on one hand, the chosen parameters correspond to a wider bandwidth for UC2; on the other hand, considering that the influence of the A0 mode cannot be completely avoided in the experiment, the selected parameters correspond to a broader full-polarization bandgap range. These results confirm that precise regulation of the bandgap's center frequency and width can be achieved by adjusting geometric parameters, endowing our metamaterial structure with tunable flexibility to adapt to varied detection application requirements.

## 3. Time domain analysis

### 3.1. Numerical simulation methods

In this section, time domain analysis is performed using Abaqus to verify the filtering capabilities of the designed MFs. First, finite element models are created to analyze the filtering capabilities of MF1 and MF2 separately, as shown in Fig. 8(a) and 8(b). A transverse displacement BC with a peak value of $1 \times 10^{-7}$ m is applied to the left side of the aluminum plate, and a 20-cycle Hamming windowed tone burst with a center frequency of 180 kHz is adopted as the excitation signal, which can be expressed as:

$$u(t) = \left[1 - \cos\left(\frac{2\pi f_c t}{n}\right)\right]\sin(2\pi f_c t) \tag{5}$$

Where $f_c$, $n$, and $t$ represent center frequency, the cycle number, and the time of the signal, respectively. In the MF1 model, a localized nonlinearity was introduced at the excitation end to mimic nonlinear interference arising from the excitation system or boundary contacts. MF1 was embedded along the propagation path to remove unwanted interference from the propagating wavefield. In the MF2 model, structural damage was introduced after excitation to simulate actual defects, with MF2 functioning to suppress the fundamental frequency component at the reception end and highlight the nonlinear harmonic response. Fig. 8(c) presents the Actual MF2 model in the Abaqus simulations, with the other models being similar to it. The aluminum plate had a thickness of 2 mm and a total length of 1200 mm. Triangular patches were periodically attached to both the upper and lower surfaces within the 568–600 mm region to form MF2. Probes were placed at 40 mm intervals along the plate to extract time domain signals, and a 200 mm damping layer was applied at the right end to eliminate reflection interference. Finally, MF1 and MF2 were used in combination to verify the proposed bidirectional guided-wave purification strategy, as shown in Fig. 8(d).

To simulate the propagation response of NGWs in the structure, particularly the superposition of system-induced and damage-induced nonlinear components, the aluminum plate was defined as a nonlinear material domain. The Landau–Lifshitz model was employed to characterize the material's nonlinear behavior [15]. This model has been widely used to describe the mechanism by which nonlinear sources such as adhesive layers and defect interfaces excite second harmonics in guided waves. Its stress–strain relationship can be expressed as:

$$\begin{aligned}\mathbf{T}^{\mathrm{RR}} & = \lambda tr[\mathbf{E}]\mathbf{I} + 2\mu\mathbf{E} + \overline{A}\mathbf{E}^2 + \overline{B}tr\left[\mathbf{E}^2\right]\mathbf{I} \\ & +2\overline{B}tr[\mathbf{E}]\mathbf{E} + \overline{C}(tr[\mathbf{E}]\,)^2\mathbf{I}\end{aligned} \tag{6}$$

Here, $\lambda$ and $\mu$ are the LamÉ constants, defined as $\lambda\& = \dfrac{E\nu}{(1+\nu)(1-2\nu)}$ and $\mu = \frac{E}{2(1+\nu)}$, where $E$ is the elastic modulus and $\nu$

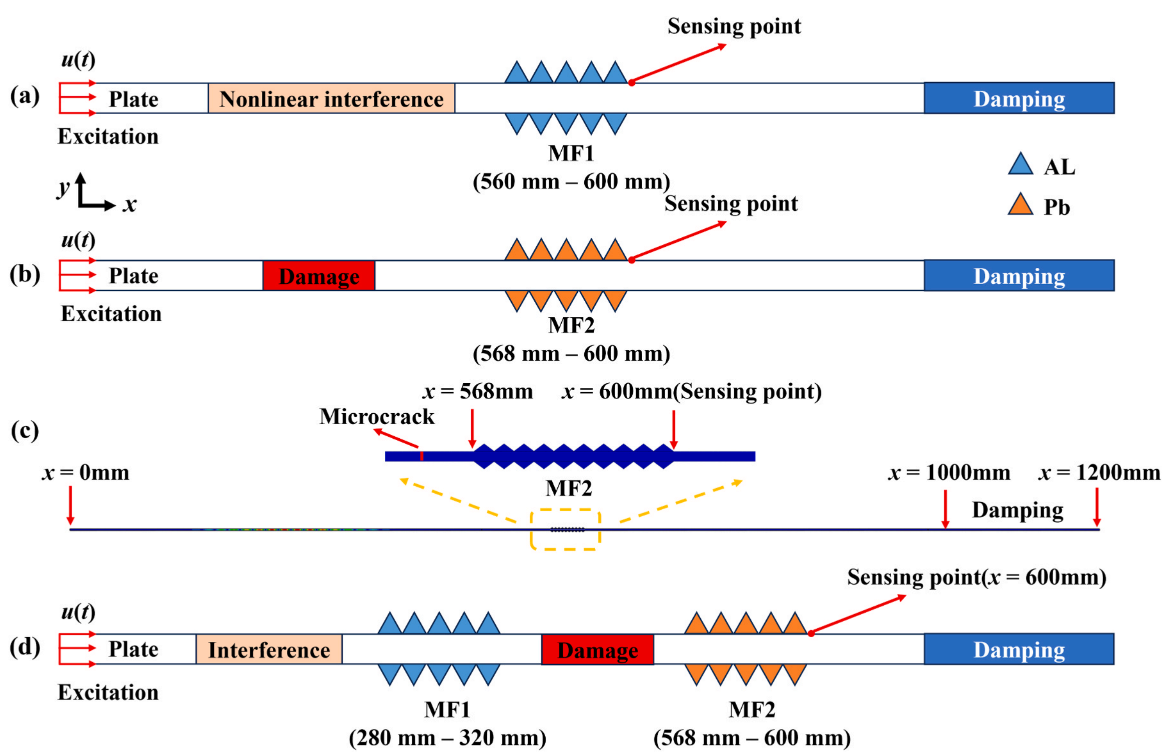


**Fig. 8.** Abaqus simulation model. (a) and (b) Simulation schematics for verifying the filtering performance of MF1 and MF2, respectively. (c) Schematic diagram of the actual MF2 model. (d) Schematic diagram of the simulation used to verify the bidirectional guided wave purification strategy.

is Poisson's ratio. $\overline{A}$, $\overline{B}$, and $\overline{C}$ are the third-order Landau constants. **I** is the second-order identity tensor, and $tr$[] represents the trace. $\mathbf{T}^{RR}$ denotes the second Piola-Kirchhoff stress tensor. **E** is the Lagrangian strain tensor, defined in terms of the displacement gradient, $\mathbf{H} = \nabla\mathbf{u}$, **as**

$$\mathbf{E} = \frac{1}{2}\left(\mathbf{H} + \mathbf{H}^{T} + \mathbf{H}^{T}\mathbf{H}\right) \tag{7}$$

The constitutive relation was implemented into the finite element simulations via a user-defined subroutine (UMAT), enabling nonlinear response within designated regions of the system. Table 2 summarizes the material parameters used in the simulations, including the properties of the adhesive, the elastic modulus of aluminum, and the Landau constants. To ensure the accuracy of the calculations, the mesh size was set to 0.1 mm and the time step to 1e −9 s

### 3.2. Results and discussion

The ability of MF1 to suppress Interfering second harmonics generated by system nonlinearities at the excitation end was first examined through time domain simulations of nonlinear guided-wave propagation with MF1. According to the setup shown in Fig. 8(a), the excitation signal is a 20-cycle Hanning windowed tone burst, as shown in Fig. 9(a), and its corresponding spectrum is shown in Fig. 9(b). To assess the influence of MF1 on fundamental frequency transmission, the excitation signal at 0 mm and the time domain signal after passing through MF1 at 600 mm were compared, as shown in Fig. 9(c), with the results indicating that MF1 has virtually no effect on the fundamental wave. Further FFT analyses were performed for probe signals at 480, 560, 600, and 680 mm, as shown in Fig. 9(d), where the fundamental component remained stable, while the second harmonic exhibited a pronounced variation during propagation. Before entering the MF1 region, the second harmonic exhibited an accumulation effect and gradually increased in amplitude. After passing through the MF1 region (560–600 mm), the second harmonic amplitude dropped sharply by about 50 times (green vs. yellow curves) and then resumed accumulation. To verify the capability of MF1 in filtering out nonlinear interference, a microcrack was introduced in front of MF1 (480 mm) in the simulation to emulate system-induced nonlinearity, and the corresponding spectrum was analyzed at 600 mm. As shown in Fig. 9(e), in the absence of MF1, comparing the purple and green curves, the amplitude of the second harmonic increases significantly after adding simulated nonlinear interference. With MF1 present, the second harmonic amplitude was reduced by nearly 50-fold (yellow vs. purple curves), while the fundamental frequency remained essentially unchanged. After introducing the nonlinear disturbance, a comparison between the red and orange curves shows that they nearly overlap at the second harmonic frequency, indicating that the nonlinear perturbation induced by the interference was completely suppressed by MF1. To quantitatively evaluate the suppression capability of MF1, Fig. 9(f) presents the normalized nonlinear coefficient $\beta$ along the propagation path, defined as

$$\beta = \frac{{A_2}^2}{{A_0}^2} \tag{8}$$

where $A_2$ and $A_0$ are the amplitudes of the second harmonic and the fundamental frequency, respectively. As shown by the blue curve in Fig. 9(f), $\beta$ increases rapidly after excitation without MF1, reflecting cumulative growth dominated by material nonlinearities. When a local nonlinear interference was introduced at 480 mm, $\beta$ exhibited a sharp increase at that location (purple curve). With MF1 present, $\beta$ dropped to nearly zero after the MF1 regardless of whether interference was introduced and then started to accumulate again from zero. This demonstrates that Interfering nonlinearities are effectively removed. Therefore, MF1 is capable of significantly suppressing Interfering

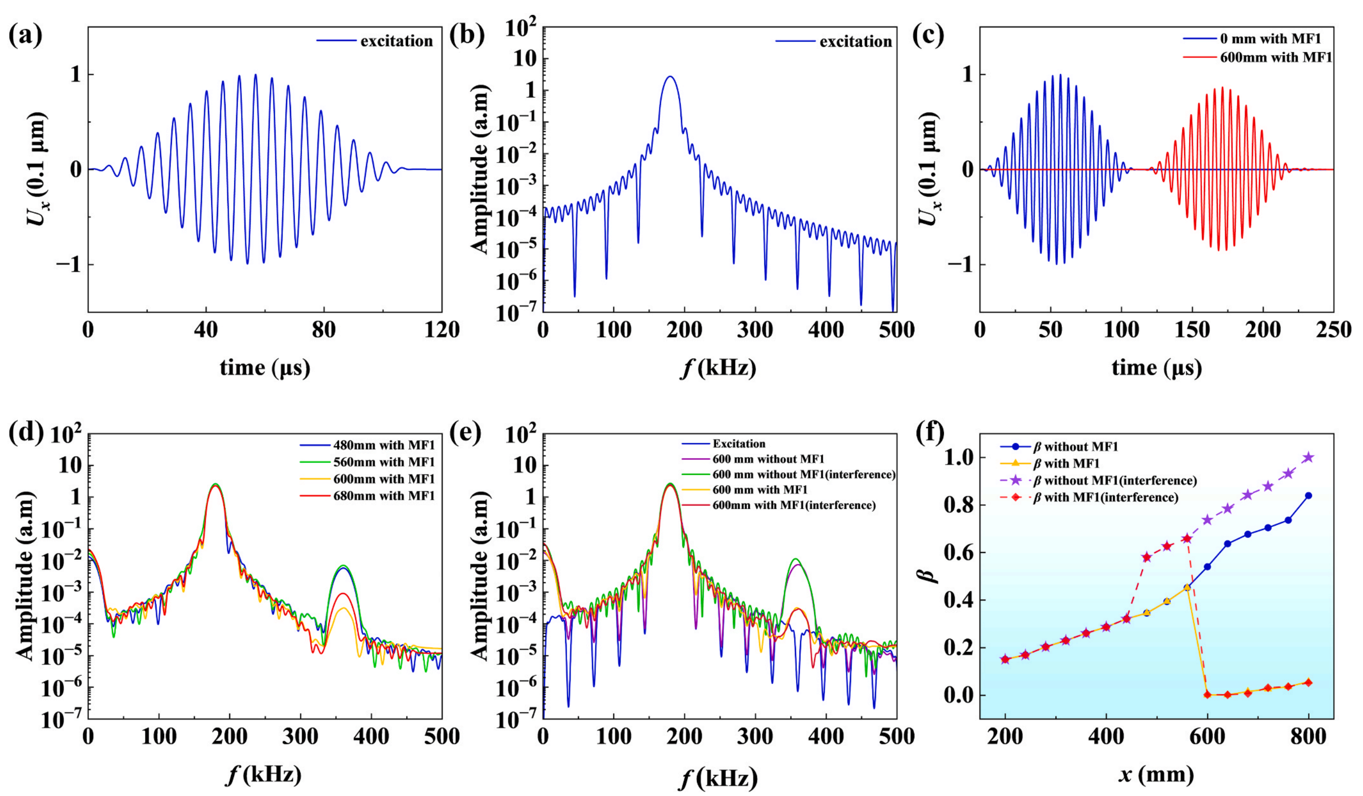


**Fig. 9.** Simulation results verifying the filtering performance of MF1. (a) Time domain waveform of the excitation signal. (b) Spectrum of the excitation signal. (c) Comparison of time domain signals at the 0 mm and 600 mm with MF1 installed. (d) Frequency domain signal comparisons at different positions (480, 560, 600, and 680 mm) with MF1 installed. (e) Frequency domain signal comparisons with and without MF1, and with or without nonlinear interference. (f) Evolution of the nonlinear coefficient $\beta$ as a function of propagation distance.

nonlinear responses originating from the excitation end without affecting the transmission of the fundamental guided wave, thereby improving both the SNR and the robustness of the detected signals in subsequent nonlinear damage identification.

To provide a more intuitive characterization of the filtering performance of MF1, propagating signals were extracted from the 600–640 mm region. Probes were placed at intervals of 0.1 mm according to the mesh size, and the signals were transformed into the frequency wavenumber domain using 2D FFT to simultaneously analyze frequency selectivity and modal propagation characteristics. During data processing, a 320–400 kHz band pass filter was applied to the signals to highlight the second harmonic components and facilitate observation of the suppression effect of MF1. As shown in Fig. 10(a), without MF1, pronounced second harmonic components were present, with energy concentrated around the S0 dispersion curve (black line), confirming that the excited mode was S0. In contrast, with MF1 installed, the nonlinear second harmonic components induced by the material have essentially disappeared, with the energy reduced to nearly zero as shown in Fig. 10(b). This observation is consistent with the result in Fig. 9(f), where the nonlinear coefficient abruptly dropped to zero at 600 mm, thereby confirming the strong filtering capability of MF1. Furthermore, when additional nonlinear interference was introduced, the second harmonic amplitude remained very low in Fig. 10(c). This demonstrates that MF1 effectively eliminates spurious interference generated by the excitation system, allowing the second harmonic to resume accumulation after MF1 and preventing damage induced nonlinear signals from being masked and rendered unidentifiable.

After validating the filtering performance of MF1, the purification capabilities of MF2 will be further analyzed. Finite element simulations were again carried out using Abaqus, with MF2 placed in the 568–600 mm region as shown in Fig. 8(b) and (c). As shown in Fig. 11(a), comparing the excitation signal with the signal after MF2 reveals that the filter completely eliminated the fundamental component. Since the nonlinear energy is much weaker than that of the fundamental wave, the signals were locally magnified to highlight their differences. As the S0 mode accumulates both second harmonic and zero frequency components during propagation, only these two components remain after MF2. Fig. 11(b) and 11(c) show the magnified waveforms for the undamaged and damaged cases, respectively, both corresponding to the superposition of zero frequency and second harmonic responses. Owing to the pronounced difference in wavelength between the zero-frequency component and the second harmonic, the two components can be effectively distinguished. The second harmonic amplitude, used as a damage indicator, increased from $\Delta A_1 = 0.0095$ (0.1 μm) in the undamaged case to $\Delta A_2 = 0.0108$ (0.1 μm) after damage was introduced. These results indicate that the structural health state can be directly assessed from the time domain signals. To further confirm this conclusion, frequency domain analysis was performed, as shown in Fig. 11(d). The results reveal that the spectral peak at the second harmonic frequency was significantly higher under the damaged condition compared with the undamaged case. This demonstrates that MF2 not only removes the dominant fundamental component with an amplitude large enough to mask the much weaker nonlinear response, but also preserves the damage-induced second harmonics, thereby enabling accurate extraction and amplification of nonlinear guided-wave signals.

To more intuitively evaluate the selectivity of MF2, signals were extracted from the 600 to 640 mm region, and frequency wavenumber spectra were computed to identify the frequency components and verify the corresponding propagation mode. Without MF2, the strongest energy was concentrated near the $S_0$ dispersion curve (black line) in Fig. 12(a), corresponding to the fundamental frequency component. When MF2 was installed, the bandgap filter effectively suppressed the fundamental frequency, allowing only the second harmonic and zero frequency nonlinear components to pass through, as seen in the two bright spots at zero frequency and 360 kHz in Fig. 12(b). When MF2 was present and damage was introduced, a significant energy enhancement appeared near $2f$ along the S0 dispersion curve in Fig. 12(c), compared to Fig. 12(b). This trend was consistent with the comparison of the time domain signals in Fig. 12(b) and (c). These results validate the functional role of MF2: suppressing the strong fundamental at the reception end while highlighting damage-induced second harmonics, thereby enhancing the contrast of nonlinear signals and reducing reliance on complex post-processing.

After separately verifying the filtering performance of MF1 and MF2, we combined them to validate our proposed bidirectional guided-wave purification strategy. Fig. 8(d) illustrates the overall filters configuration: MF1 was placed the excitation end (280–320 mm) to suppress second harmonics interference, while MF2 was installed the reception end (568–600 mm) to filter out the strong fundamental wave that masks the second harmonic. To evaluate the structural function, interference and damage were introduced separately, and the in-plane displacements after passing through MF1 and MF2 were extracted at 600 mm for comparison. As shown by the blue curve in Fig. 13(a), the second harmonic amplitude was $\Delta A_3 = 0.0037$ (0.1 μm) when no disturbance or damage was introduced. Next, we introduced a nonlinear interference at 200 mm before MF1. The time domain signal at 600 mm is shown by the yellow curve in Fig. 13(a), where the second harmonic amplitude is $\Delta A_4 = \Delta A_3 = 0.0037$ (0.1 μm), indicating that MF1 effectively suppressed the second harmonic interference. Then, we introduced damage at 480 mm between MF1 and MF2. The second harmonic amplitude of the corresponding red signal in Fig. 13(a) significantly increased to $\Delta A_5 = 0.0092$ (0.1 μm), reflecting the enhanced nonlinear response caused by the interaction between the damage and the wave. Fig. 13(b) shows the spectrum corresponding to (a). Observing the blue and yellow lines, it can be seen that when there is no damage, the signals with and without introduced interference exhibit the same second harmonic

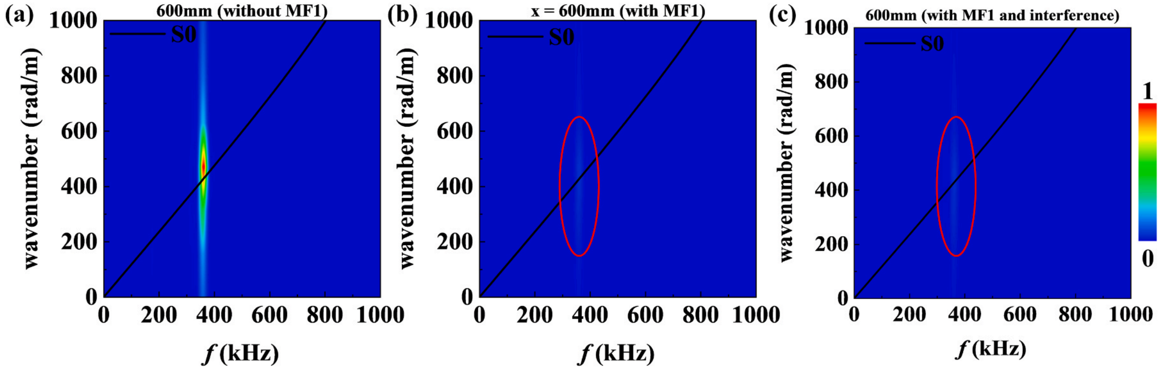


**Fig. 10.** Frequency wavenumber spectra obtained via 2D FFT of the band pass filtered signals extracted after MF1 (600–640 mm). (a) Without MF1. (b) With MF1 installed. (c) With MF1 and interference.

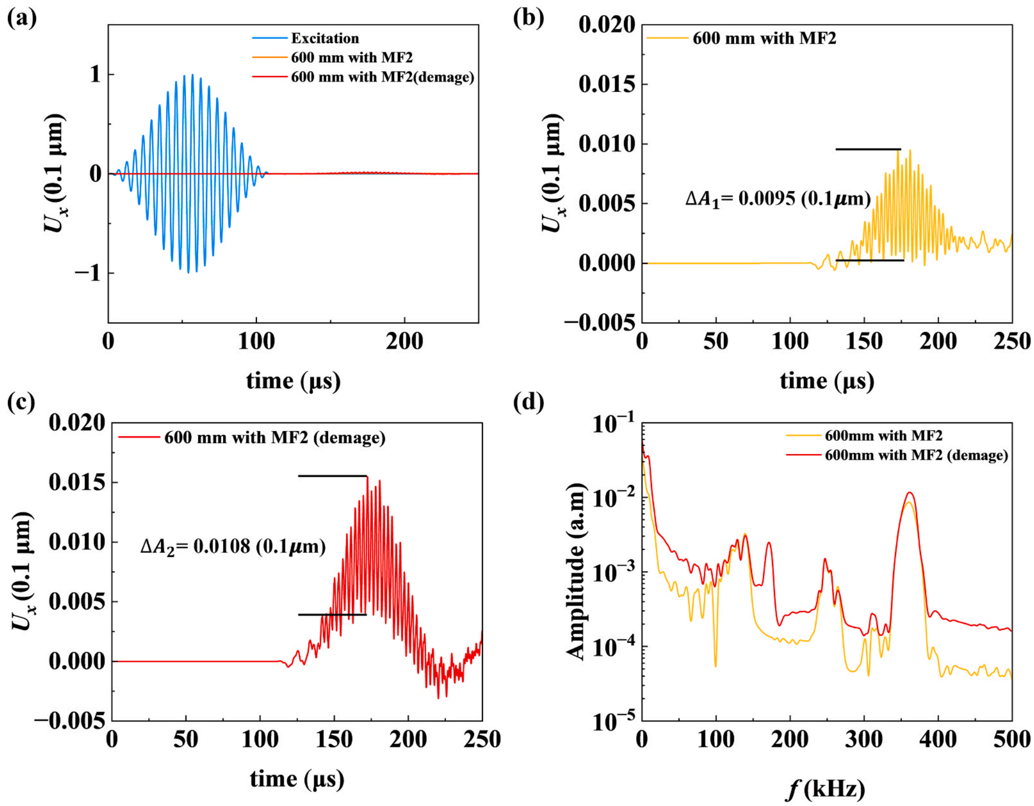


**Fig. 11.** Simulation results verifying the filtering performance of MF2. (a) Comparison between the excitation signal and the signal after propagating through MF2 at 600 mm. (b) Time domain signal at 600 mm with MF2 installed, without damage. (c) Time domain signal at 600 mm with MF2 installed, with damage. (d) Frequency domain comparison of signals received at 600 mm with and without damage when MF2 is present.

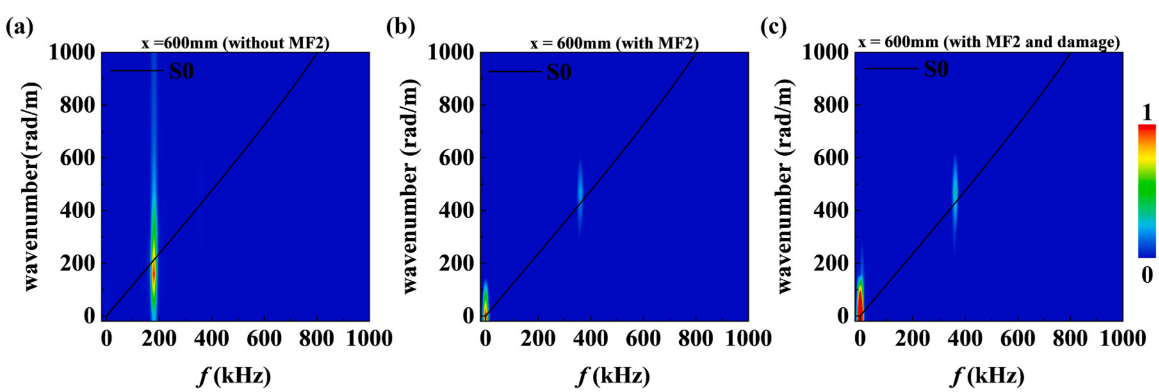


**Fig. 12.** Frequency wavenumber spectra obtained via 2D FFT of signals extracted after MF2 (600–640 mm). (a) Without MF2. (b) With MF2 installed. (c) With MF2 and damage.

component. However, after introducing damage, the second harmonic component of the red line segment is significantly enhanced, making the second harmonic measurement more intuitive. This demonstrates that our designed filters configuration not only suppresses system-induced nonlinear interference but also significantly amplifies damage-induced second harmonics at the receiver end. Consequently, structural health status can be directly identified from the time domain signal. In summary, the combined use of MF1 and MF2 simultaneously suppresses multi-source nonlinear interference and enhances the detectability of damage-induced nonlinear components, thereby reducing the complexity of subsequent signal processing.

## 4. Experimental investigation

### 4.1. Experimental setup

Based on the results obtained from the numerical investigations, an experimental study was conducted under laboratory conditions to evaluate the performance of the MFs and to assess their potential applicability in practical scenarios. The experimental setup was carefully designed to closely replicate the conditions considered in the numerical models. The purification capability of MF1 and MF2 was systematically examined, aiming to experimentally validate the feasibility of the proposed bidirectional guided-wave purification strategy. Fig. 14(a) and (b) present the schematic illustration and the photograph of the experimental setup, respectively. The specimen consisted of an

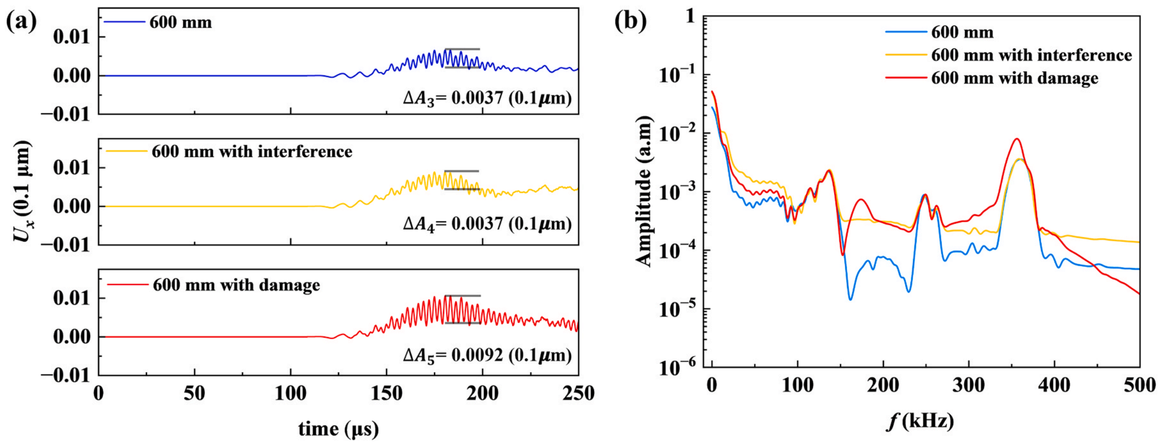


**Fig. 13.** Simulation verification with MF1 and MF2 installed simultaneously. (a) Time domain signals collected at 600 mm, comparing the signals with those obtained under interference and damage. (c) Comparison of frequency domain signals.

aluminum plate fabricated using laser cutting technology with a machining precision of 0.1 mm. The plate had a length of 1200 mm, a width of 12 mm, and a uniform thickness of 2 mm, and was made of 6061 aluminum alloy. Both MF1 and MF2 were composed of ten periodically arranged UC1 and UC2 units, respectively. Their geometric dimensions were identical to those used in the numerical simulations. The constituent materials of MF1 and MF2 were 6061 aluminum alloy and lead, respectively, as shown in Fig. 14(c)–(e). To ensure stable support of the specimen and to avoid adverse effects caused by direct contact with the platform, the aluminum plate was placed on two rubber pads. In addition, the right end of the plate was wrapped with BluTack to form a damping layer, so as to minimize the influence of boundary reflections on the measured signals. Guided Lamb waves were generated by a square Lead Zirconate Titanate (PZT) transducer with dimensions of 12 mm × 12 mm × 1 mm, which was bonded to the left end of the plate using 3 M VHB adhesive. The PZT transducer was driven by a signal generator (Tektronix AFG 31102) in conjunction with a power amplifier (Aigtek ATA-214). To produce sufficiently strong nonlinear responses for validating the functionality of the MFs, a high-power 50-cycle Hanning windowed tone burst was employed as the excitation. The designed MFs were bonded to both the upper and lower surfaces of the aluminum plate using UHU 2-component epoxy, with their installation locations consistent with those in the numerical model. The plane vibration response at the monitoring point was measured using a laser vibrometer sensor (SOPTOP LV-S01). The measured signals were synchronously transmitted to an oscilloscope through the laser measurement system for real-time monitoring, ensuring synchronization between the PZT excitation and the displacement measurements.

The experimental procedure followed the same sequence as that adopted in the numerical simulations. First, MF1 and MF2 were installed separately to calculate their transmission coefficients and to verify their individual filtering performances. Subsequently, MF1 and MF2 were installed simultaneously, and the guided-wave responses were measured under conditions with introduced nonlinear interference and damage, in order to further validate the effectiveness of the proposed bidirectional guided-wave purification strategy. Previous studies have demonstrated that a cylinder coupled to a plate through a gel type couplant can act as a nonlinear scatterer and generate pronounced higher order harmonic components [48]. Such enhancement of nonlinear components is mainly attributed to contact nonlinear effects at the plate–cylinder interface. Based on this observation, a cylindrical object with a diameter of 10 mm and a thickness of 12 mm was employed in this study to model both mock-up interference and damage, as illustrated in Fig. 14(f) and (g).

### *4.2. Results and discussion*

First, the capability of MF1 to purify nonlinear interference was experimentally investigated. In the experiments, guided-wave responses of the aluminum plate were measured under two conditions: without MF1 and with MF1 bonded in the region from 280 to 320 mm. A monitoring point was placed at 360 mm to compare the influence of MF1 on guided-wave propagation. To characterize the frequency selective behavior of MF1, its transmission coefficient was measured using a linear chirp signal covering a frequency range from 5 kHz to 1000 kHz with a step size of 100 Hz. The frequency spectra of the monitored signals obtained with and without MF1 were taken as $u_1$ and $u_2$ in Eq. (1), respectively, and the calculated transmission coefficient is shown in Fig. 15. A pronounced bandgap can be clearly observed around 360 kHz, indicating that MF1 exhibits a strong attenuation effect on guided waves within this frequency range. This experimental observation is in good agreement with the numerical predictions.

Based on these results, 50-cycle tone burst signals with center frequencies of 180 kHz and 360 kHz were further employed to validate the linear filtering performance of MF1. Fig. 16(a) and (b) present the comparison of normalized time domain signals measured at the monitoring point under 180 kHz excitation, without and with MF1. It can be observed that the 180 kHz signal lies within the passband of MF1 and can be transmitted through MF1 with high efficiency. When the excitation center frequency was set to 360 kHz, as shown in Fig. 16(c) and (d), the monitored signal amplitude was significantly reduced because this frequency falls within the bandgap of MF1, indicating that the second harmonic frequency component was effectively suppressed.

The results confirm that MF1 can effectively filter out energy in the bandgap frequency range while maintaining high transmittance of the fundamental wave, consistent with its expected filtering and purification functions.

To further verify the selective filtering characteristic of MF1 from the frequency domain perspective, spectral analyses were performed on the time domain signals shown in Fig. 16(a) and (b), and the results are presented in Fig. 17. It is clearly observed that the spectral amplitude around 360 kHz is substantially lower when MF1 is installed compared with the case without MF1, whereas the fundamental frequency component remains at a relatively high level. These results demonstrate that MF1 is capable of effectively suppressing nonlinear high frequency components within a specific frequency range without affecting the propagation of the fundamental guided wave, thereby exhibiting effective nonlinear interference purification performance.

After the individual verification of the purification capability of MF1, further experimental validation was carried out to evaluate the functionality of the receiving-end MF2. In the experiments, MF2 was bonded

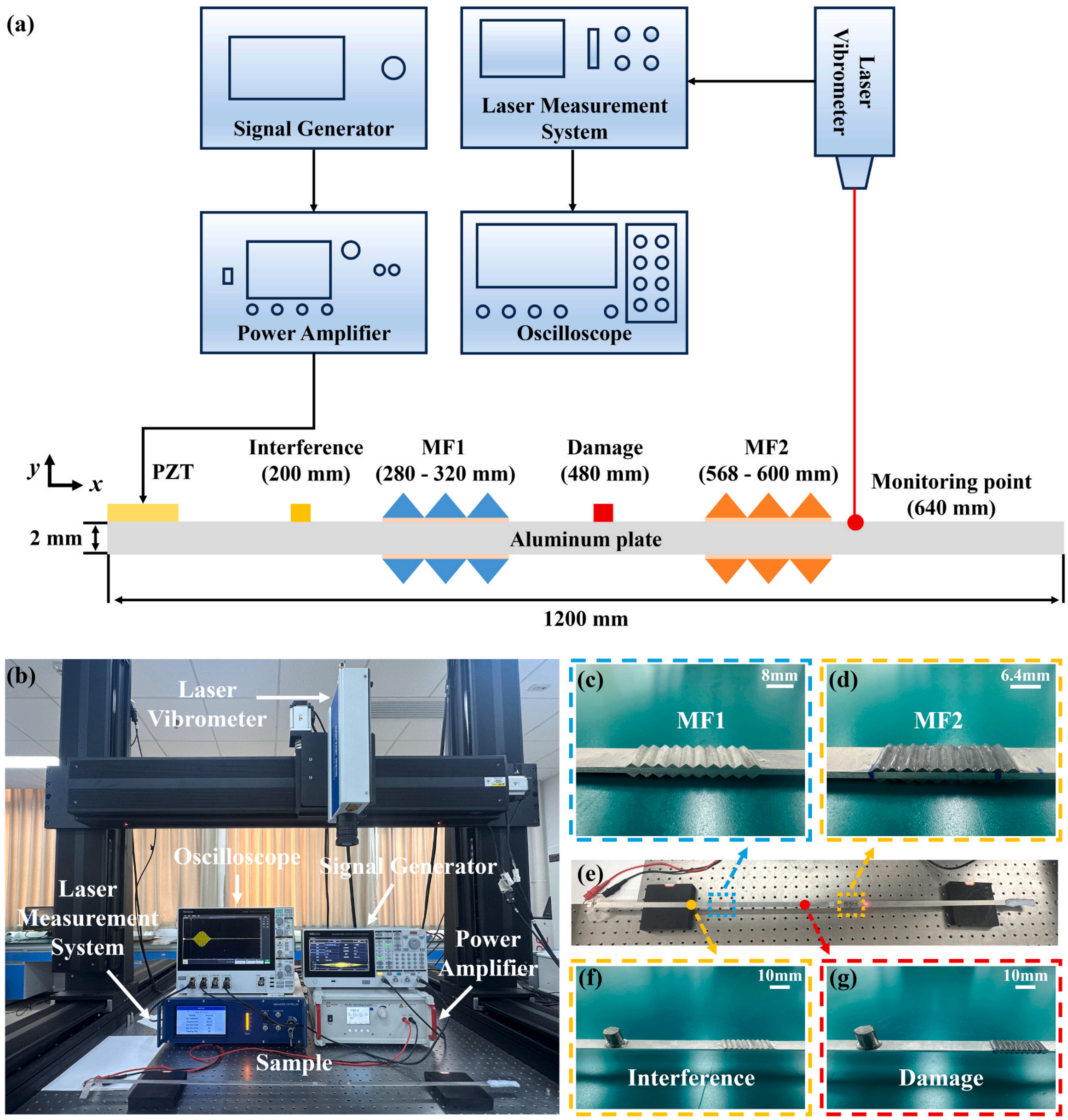


**Fig. 14.** (a) Schematic and (b) experimental setup for evaluating the performance of MFs in nonlinear guided wave testing. (c) Close-up view of MF1. (d) Close-up view of MF2. (e) Sample. (f) Mock-up interference. (g) Mock-up damage.

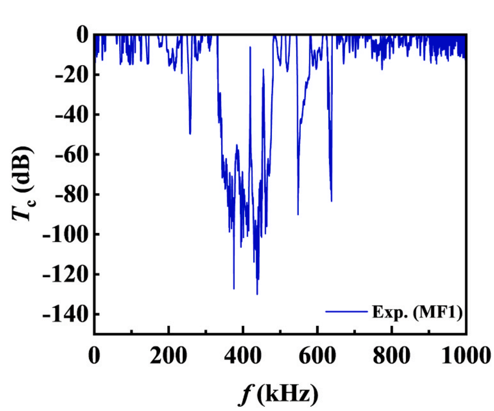


**Fig. 15.** Experimentally measured MF1 transmission coefficient.

to the aluminum plate in the region from 568 to 600 mm, and a monitoring point was placed at a distance of 640 mm from the excitation end to measure the received signals. The validation procedure for MF2 was generally consistent with that adopted for MF1. First, the transmission coefficient of MF2 was measured using a linear chirp excitation with a frequency range of 5 kHz to 600 kHz, and the experimental results are shown in Fig. 18. In contrast to the filtering behavior of MF1, MF2 exhibits a pronounced bandgap around 180 kHz, while operating in the passband near 360 kHz. This frequency selectivity capability indicates that MF2 is capable of effectively suppressing the fundamental frequency component at the receiving end, thereby eliminating its interference with the observation of the second harmonic response.

To further verify the linear filtering performance of MF2, tone burst signals with center frequencies of 180 kHz and 360 kHz were employed as excitations. Figs.19(a) and (b) present the comparison of normalized time domain signals measured at 640 mm under 180 kHz excitation, without and with MF2. It can be observed that the amplitude of the fundamental frequency component is significantly reduced after MF2 is installed, demonstrating the effective suppression of the 180 kHz guided

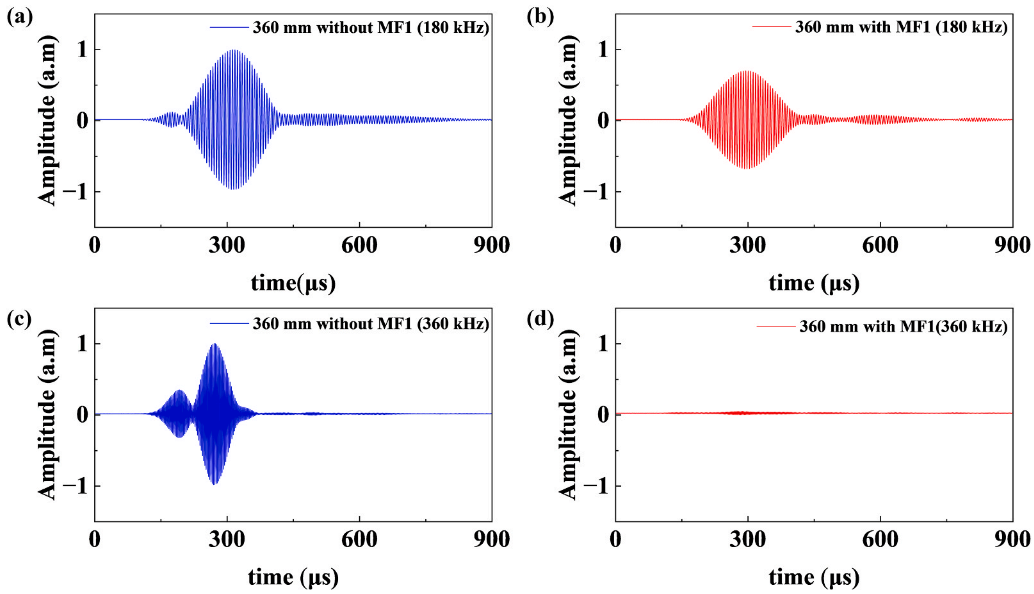


**Fig. 16.** Experimentally measured signal at 360 mm under 180 kHz excitation: (a) without MF1; (b) with MF1. Experimentally measured signal at 360 mm under 360 kHz excitation: (c) without MF1; (d) with MF1.

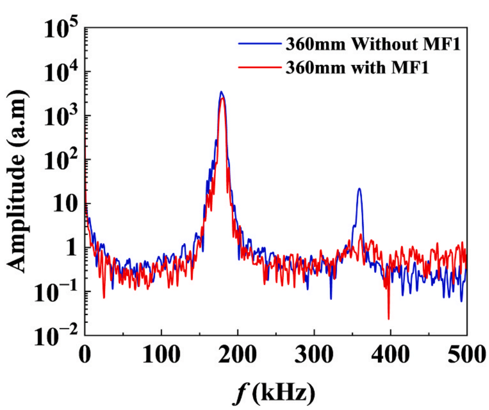


**Fig. 17.** Experimentally measured frequency domain signal at 360 mm without and with MF1.

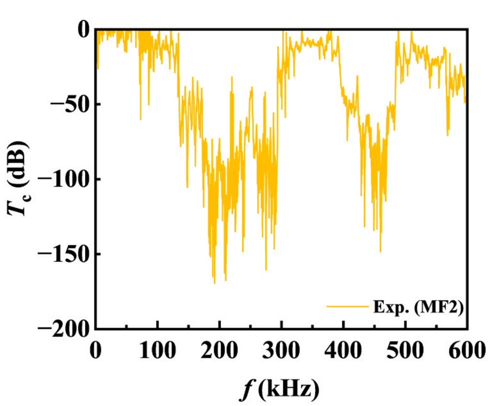


**Fig. 18.** Experimentally measured MF2 transmission coefficient.

wave by MF2. Figs.19(c) and (d) show the corresponding normalized time domain signals measured at 640 mm under 360 kHz excitation. After the installation of MF2, the 360 kHz signal remains clearly observable, indicating that MF2 allows efficient transmission of second harmonic in this frequency range. These results confirm the pronounced frequency selective filtering capability of MF2 and establish a solid experimental basis for real-time monitoring of second harmonic components at the receiving end.

It should be noted that, because the energy of the fundamental frequency component is much higher than that of the second harmonic component, the latter cannot be directly observed under the same amplitude scale. Therefore, the vertical axis range of the signal in Fig. 19 (b) was adjusted, and the waveform was magnified on the oscilloscope, yielding a clearly identifiable second harmonic signal with a peak-to-peak amplitude of $\Delta A_6 = 0.004$, as shown in Fig. 20(a). To further confirm that the observed signal originates from the second harmonic component, a FFT was applied to the time domain signal, and the resulting frequency domain spectrum is presented in Fig. 20(b). It can be seen that the fundamental frequency component has been effectively suppressed by MF2, and only the second harmonic component at 360 kHz remains in the spectrum, thereby verifying the effectiveness of MF2 in suppressing the fundamental wave and extracting the second harmonic response at the receiving end.

After individually verifying the functionality of MF1 and MF2, MF1 and MF2 were installed simultaneously on the aluminum plate at the locations illustrated in Fig. 14(a) to experimentally validate the proposed bidirectional guided-wave purification strategy. This strategy aims to suppress nonlinear interference while enabling the damage-induced nonlinear components to be directly observed on the oscilloscope. With MF1 and MF2 installed simultaneously, the time domain signal measured at a distance of 640 mm from the excitation end is shown by the blue curve in Fig. 21(a). Under this intact condition without interference or damage, the peak-to-peak amplitude of the second harmonic component was $\Delta A_6 = 0.0014$, representing the baseline nonlinear response of the system. Subsequently, a mock-up interference source was introduced approximately 200 mm upstream of MF1 to generate system-induced nonlinear disturbances, as shown in Fig. 14(f), to examine the capability of MF1 in purifying nonlinear interference. The corresponding measurement result is indicated by the yellow curve in Fig. 21(a). It can be observed that, after introducing the mock-up interference, the amplitude of the second harmonic signal exhibited almost no increase compared with the intact case, with a peak-to-peak value of $\Delta A_7 = 0.0015$, demonstrating the effective suppression of nonlinear interference by MF1. Next, a mock-up damage was introduced at approximately 480 mm between MF1 and MF2, and the

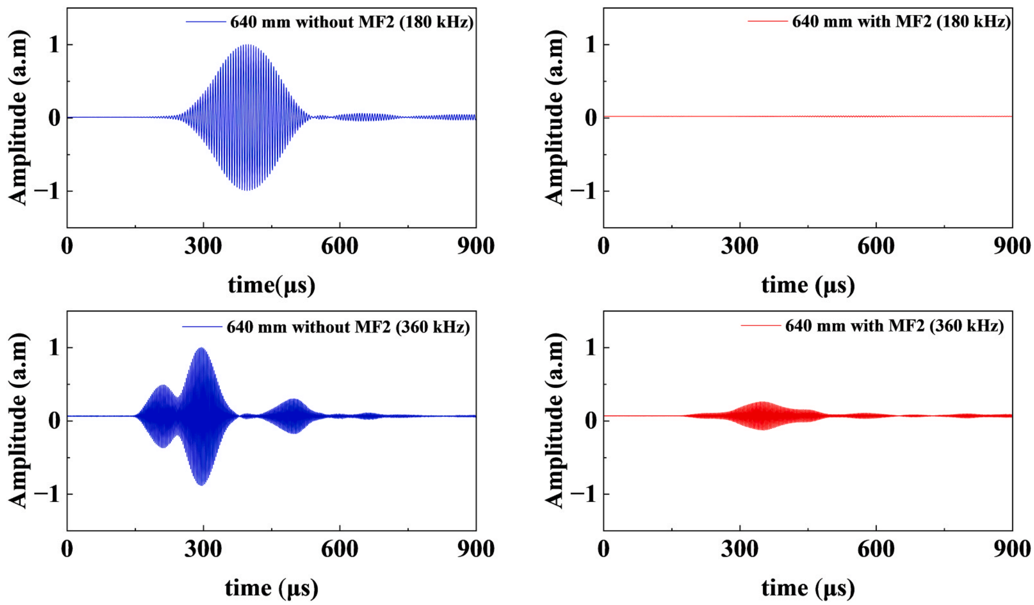


**Fig. 19.** Experimentally measured signal at 640 mm under 180 kHz excitation: (a) without MF2; (b) with MF2. Experimentally measured signal at 360 mm under 360 kHz excitation: (c) without MF2; (d) with MF2.

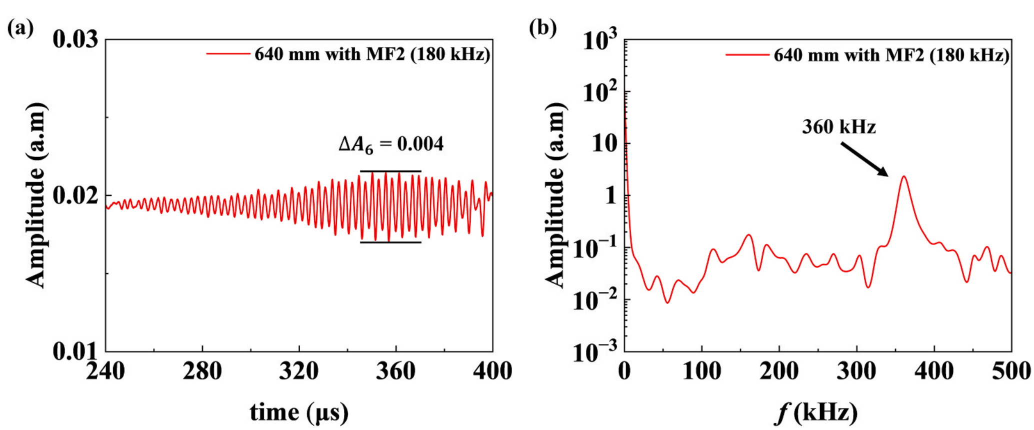


**Fig. 20.** Experimentally measured signals at 640 mm under 180 kHz excitation with MF2 installed: (a) time domain; (b) frequency domain.

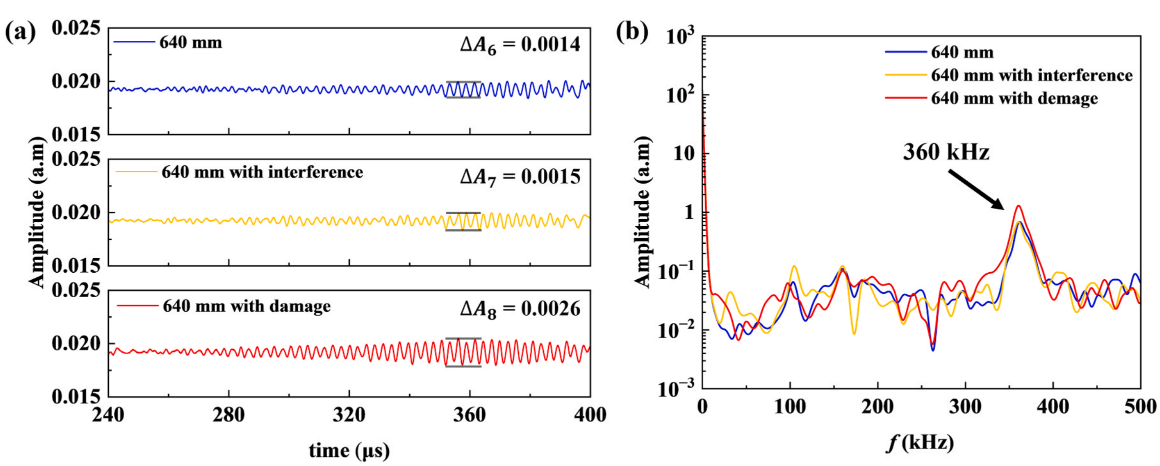


**Fig. 21.** Experimentally measured signals at 640 mm with MF1 and MF2 installed simultaneously: (a) time domain and (b) frequency domain. Results under three conditions are compared, including the intact case, the case with mock-up interference, and the case with mock-up damage.

variation of the measured signal was examined. The resulting time domain signal is shown by the red curve in Fig. 21(a), where a pronounced increase in the second harmonic amplitude can be clearly observed. The corresponding peak-to-peak value increased to $\Delta A_8 = 0.0026$, indicating a significantly enhanced nonlinear response induced by the damage.

Fig. 21(b) presents the frequency domain spectra corresponding to the three experimental conditions. It can be seen that MF2 effectively suppresses the fundamental frequency component that would otherwise mask the second harmonic response, leaving the spectral content dominated by the second harmonic component around 360 kHz. By comparing the blue and yellow curves, it is evident that the introduction of mock-up interference upstream of MF1 does not lead to an increase in the second harmonic amplitude. In contrast, when mock-up damage is introduced between MF1 and MF2, the second harmonic component exhibits a clear enhancement, as shown in Fig. 21(b). These frequency domain observations are consistent with the time domain results discussed above.

The above experimental results further confirm the effectiveness of the proposed MF system. The system not only prevents damage-induced nonlinear signals from being masked by nonlinear components generated by system interference, but also enables the real-time identification of the structural health condition through variations in the second harmonic time domain signal, thereby significantly reducing the complexity of subsequent signal processing and nonlinear feature extraction. This study provides a new experimental strategy for real-time damage monitoring in SHM systems based on NGWs, and demonstrates the practical potential of metamaterial-assisted designs in realistic SHM applications. The proposed approach lays a solid foundation for the future development of metamaterial-assisted SHM technologies.

## 5. Conclusions

This study proposes and systematically investigates a nonlinear guided-wave purification method based on bidirectional MFs, demonstrating the potential of metamaterial-assisted designs for enhancing the real-time detection of early structural damage using NGWs. By bonding specially designed locally resonant bandgap metamaterials onto the surface of the structure, undesired system-induced nonlinear interference can be effectively suppressed at the excitation end, thereby improving the accuracy and reliability of second harmonic-based SHM. Meanwhile, the second harmonic component can be selectively extracted at the receiving end, enabling real-time identification of the structural health condition. Comprehensive theoretical and numerical analyses, including dispersion calculations as well as frequency domain and time domain investigations, confirm that the designed metamaterial bandgaps exhibit good tunability and robustness through appropriate geometric parameter control. The results show that the proposed metamaterial structures can effectively attenuate frequency components within the bandgap while allowing efficient transmission of out-of-bandgap guided waves. Corresponding experimental results further validate the selective filtering capability of the metamaterial surfaces under practical conditions. In addition, comparative experiments involving locally introduced nonlinear interference and simulated damage reveal that the proposed bidirectional MFs system significantly enhances the sensitivity of NGWs to damage evolution. Damage-related second-harmonic components can be effectively extracted and directly monitored in real time on an oscilloscope, highlighting the practical advantage of the proposed approach. Overall, this work provides a new experimental strategy for nonlinear guided-wave-based SHM systems and demonstrates the promising potential of metamaterial designs in practical SHM applications. The proposed approach lays a solid foundation for the future development of high-reliability and low-complexity metamaterial-assisted SHM technologies.

## CRediT authorship contribution statement

**Weibin Li:** Writing – original draft, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Huanyang Chen:** Supervision, Investigation, Formal analysis. **Mingxi Deng:** Supervision, Methodology, Funding acquisition, Conceptualization. **Yuhang Yin:** Software. **Yifei Xu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation.

## Author contributions

H.Y.C., M.X.D. and W.B.L. supervised the project. Y.F.X. conceived ideas for the project and performed the theoretical analysis. Y.F.X. and Y.H.Y. performed numerical simulations. Y.F.X. performed the experiments. Y.F.X. wrote the manuscript.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

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